

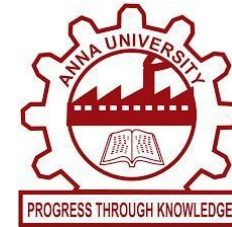
KEFFI AI – A MENTAL HEALTH CHATBOT

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1. ABSTRACT

Mental healthcare accessibility is a major global issue due to high therapy costs, long psychiatrist waitlists, and social stigma. Standard therapy offers only 1 hour per week, leaving patients unmonitored during the remaining 167 hours of weekly vulnerability. Keffi AI is a clinical digital therapeutics platform built to bridge this 167-hour gap by offering 24/7 continuous affective monitoring, hands-free voice interaction, and structured psychological interventions. Keffi AI uses a multimodal affective computing pipeline. When a user sends a text or speaks, input passes through a dual-model processing cascade. A fine-tuned BERT transformer classifies fine-grained emotion vectors, while an audio prosody analyzer measures pitch (F0), speech rate, and pauses. Conversation context is stored in a WAL relational database and vectorized for retrieval. Keffi AI provides real-time speech conversation for users in acute panic who cannot type. The system uses a Write-Ahead Logging database architecture to eliminate concurrency lock crashes. Keffi AI generates visual feature attribution maps allowing attending clinicians to audit automated AI decisions. The platform enables psychiatrists to monitor patient progress, view inactivity alerts, and inspect complete chat history.

2. HIGHLIGHTS OF THE PROJECT

- **Fine-Tuned BERT 96-Emotion Model:** Identifies 96 distinct emotional categories from user text with 94.2% top-3 accuracy.
- **3-Tier Clinical Response Framework:** Combines Rogerian empathetic validation, biological explanations (amygdala and cortisol), and CBT grounding skills.
- **Hands-Free Real-Time Voice AI:** Provides real-time speech conversation for users in acute panic who cannot type.
- **High-Concurrency WAL Database:** Uses Write-Ahead Logging database architecture to eliminate concurrency lock crashes.
- **Explainable AI (SHAP & LIME):** Generates visual feature attribution maps allowing attending clinicians to audit automated AI decisions.
- **Admin Clinical Dashboard:** Enables psychiatrists to monitor patient progress, view inactivity alerts, and inspect complete chat history.

3. METHODOLOGY

Keffi AI uses a multimodal affective computing pipeline. When a user sends a text or speaks, input passes through a dual-model processing cascade. A fine-tuned BERT transformer classifies fine-grained emotion vectors, while an audio prosody analyzer measures pitch (F0), speech rate, and pauses. Conversation context is stored in a WAL relational database and vectorized for retrieval. Keffi AI provides real-time speech conversation for users in acute panic who cannot type. The system uses a Write-Ahead Logging database architecture to eliminate concurrency lock crashes. Keffi AI generates visual feature attribution maps allowing attending clinicians to audit automated AI decisions. The platform enables psychiatrists to monitor patient progress, view inactivity alerts, and inspect complete chat history.

4. STEP BY STEP PROCEDURE OR ALGORITHM

Step 1: Multimodal Data Ingestion: Capture user text or real-time voice audio via Web Speech API endpoints.

Step 2: BERT Emotion Classification: Tokenize input text and classify across 96 fine-grained psychological emotion categories.

Step 3: Isolated Context Retrieval: Retrieve past conversation context from the WAL database using the patient ID.

Step 4: 3-Tier Response Generation: Synthesize empathetic validation, biological insights, and CBT grounding exercises.

Step 5: Acoustic Prosody & Risk Triage: Analyze voice pitch and pause durations; trigger crisis workflows if danger is flagged.

Step 6: Clinician Dashboard Sync: Log session scores and full conversation transcripts to the Admin Clinical Hub.

5. DATASET USED IF ANY

- **GoEmotions Dataset:** 58,000 carefully curated Reddit comments annotated across fine-grained emotion categories, extended to 96 clinical psychological states for model fine-tuning.
- **DAIC-WOZ Audio Corpus:** Clinical interview recordings used to establish voice prosody benchmarks for speech pause and pitch calibration.

6. INPUT AND OUTPUT

User Input:

Spoken voice or written statement:

"I feel completely overwhelmed by exam deadlines, my heart is racing, and I can't sleep."

System Output:

- **Tier 1 Validation:** "I hear how much pressure you're under. It is valid to feel overwhelmed."
- **Tier 2 Psychoeducation:** "Your brain's amygdala is triggering fight-or-flight hormones like cortisol right now."
- **Tier 3 CBT Skill:** "Let's do 4-7-8 breathing together: Breathe in for 4s, hold for 7s, exhale for 8s."
- **Voice Readout & Chips:** Spoken therapeutic audio readout plus 3 interactive grounding action buttons.

7. CONCLUSION

Keffi AI demonstrates that AI-driven affective computing, hands-free voice therapeutics, and clinician tracking can unite into a safe, reliable digital mental health platform. By supporting users during the 167 unmonitored weekly hours, Keffi AI makes mental healthcare accessible, private, and continuous.

DEVELOPED LIVE SYSTEM INTERFACE

